1. A local distributor for a national tire company expects to sell approximately 9,600 steel-belted radial tires of a certain size and tread design next year. The annual carrying cost is $16 per tire, and the ordering cost is $75. The distributor operates 288 days a year.
   a) What is the EOQ?
   b) How many times per year does the store reorder?
   c) What is the length of an order cycle?
   d) What is the total annual cost if the EOQ quantity is ordered?

2. A manufacturing company places an annual order of 48000 units at a price of $20 per unit. Its carrying cost is 15% and the order cost in 12 per order.
   a. What is the most economic order quantity?
   b. How many orders need to be placed?
   c. Total annual cost of ordering and carrying the inventory.

3. The maintenance department of a large hospital uses about 816 cases of liquid cleanser annually. Ordering costs are $12, carrying costs are $4 per case a year, and the new price schedule indicates that orders of less than 50 cases will cost $20 per case, 50 to 79 cases will cost $18 per case, 80 to 99 cases will cost $17 per case, and larger orders will cost $16 per case. Determine the optimal order quantity and the total cost.

4. A firm must meet the following demand over 4 periods:

Period (t)	1	2	3	4
Demand (D_t)	20	30	40	10

Fixed ordering/setup cost (K) = 100 per order

Holding cost (h) = 2 per unit per period (charged on ending inventory)

No shortages allowed, Initial inventory = 0

Use the Wagner–Whitin algorithm to determine the optimal ordering policy and minimum total cost.